

Test Points

Test point	Details
TP0, TP4	0V (GND)
TP1, TP5	9V DC
TP2, TP8	Data
TP3, TP6	5V DC
TP7	3.3V DC

command to the robot is sent through the RF transmitter module wirelessly.

These transmitted signals are received at the receiver end through an RF receiver module and transferred to Arduino Board1 at the receiving end, which then drives L293D IC to rotate the motors for robotic movement.

Arduino UNO board. Arduino UNO is an open source electronics prototyping platform based on flexible, easy-to-use hardware and software. Arduino UNO development board is built around ATmega328 chip. It has 14 digital input/output pins [of which six can be used as pulse-width modulation (PWM) outputs], six analogue inputs, a 16MHz ceramic resonator, USB connection, power jack, an ICSP header and reset button. It contains everything for supporting the microcontroller (MCU); simply connect it to a computer with a USB cable or power it with an AC-to-DC adaptor or battery to get started.

ADXL345 module. ADXL345 accelerometer modules are available from various manufacturers. Shapes and sizes of the modules may differ but these all have SDA, SCL, SDO and CS pins for I2C and ISP serial communications. Pin details of a typical ADXL345 module are shown in Fig. 4.

You may also use a module equivalent to ADXL345 such as Keyes-345 accelerometer module. It communicates with Arduino using I2C communication protocol. Here, Arduino UNO board acts as the master device and ADXL345 as the slave device. SDA and SCL pins of ADXL345 should be connected to the SDA pin (pin 31) and SCL pin (pin 32) on Arduino UNO board, respectively.



Fig. 4: Pin details of a typical ADXL345 module

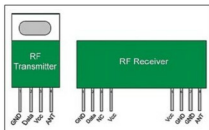


Fig. 5: RF transmitter and receiver modules

Readings of the accelerometer will give the tilt direction of ADXL345 on all three axes (X, Y and Z) by varying the gesture of the hand.

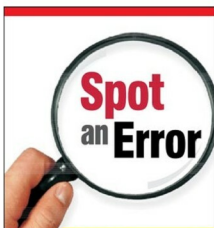
RF module. An RF module is a fast and effective device for wireless communication for shorter ranges. Here, we have used a pair of RF transmitter and receiver modules, operating at 433MHz. A library of Arduino, named VirtualWire.h, is used to sync the RF module with Arduino. Of the eight pins of the RF receiver, only four pins, namely, Antenna, 5V, GND and Data are used here. Connect a single-strand wire to Antenna pin of both transmitter and receiver pins of the RF modules as shown in Fig. 5.

Software program

Program for this robot includes Wire.h header file for I2C communication protocol and VirtualWire.h library of Arduino to sync the RF module with Arduino.

VirtualWire.h is an Arduino library that provides features to send short messages. It allows you to send and receive data bytes and strings easily.

The software has the following functions:



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